

Continual Learning for Husky Robot via Policy Distillation

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Simulation Environment

Simulation Environment: **Pybullet**

Robot: **Husky**

Action Space: linear velocity $\in [-1, 2]$

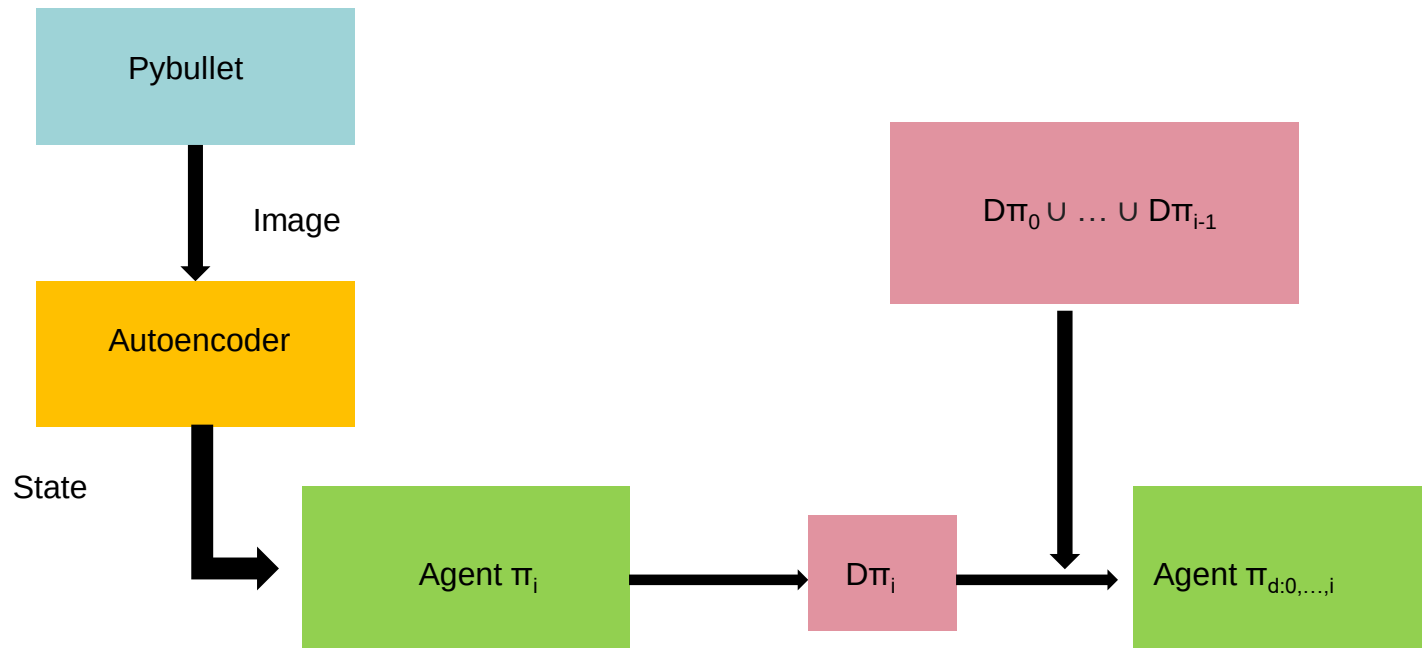
angular velocity $\in [-\pi/2, \pi/2]$

State Space: 64 variables

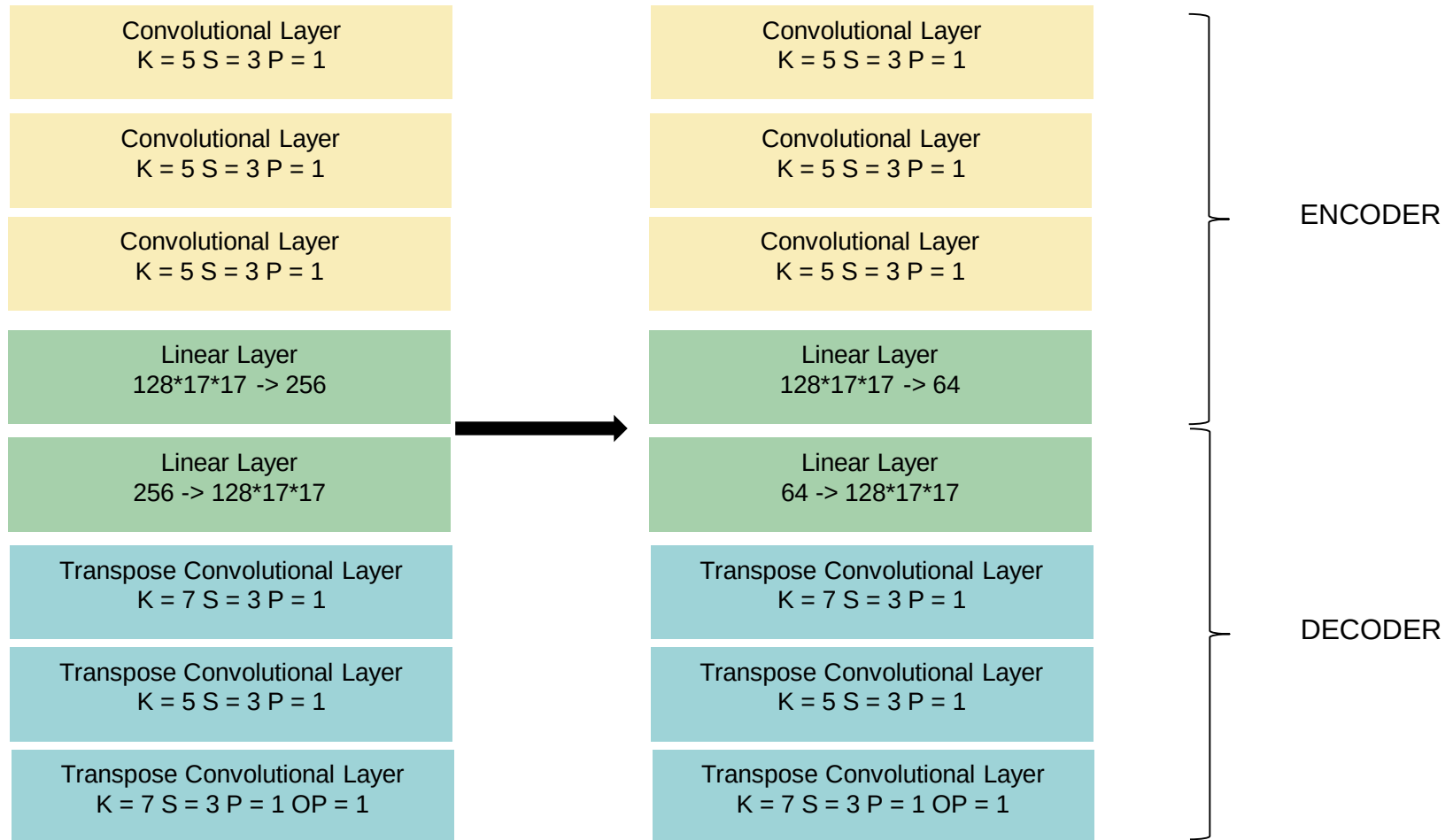
The environment is vectorized for Stable-Baseline
3 with **DummyVecEnv**



Project Pipeline



Autoencoder



First Task

The task consists in reaching a red square in the scene.
The agent has been trained for over 2M timesteps.



Reward terms:

- Reward for getting closer to the target
- Reward for reducing the vertical distance between the robot and the target
- Reward for reaching the target
- Penalty for moving too far from the center of the scene

Second Task

The task consists in circling around a blue target.
The agent has been trained for over 2M timesteps.



The reward is defined by the following formula:

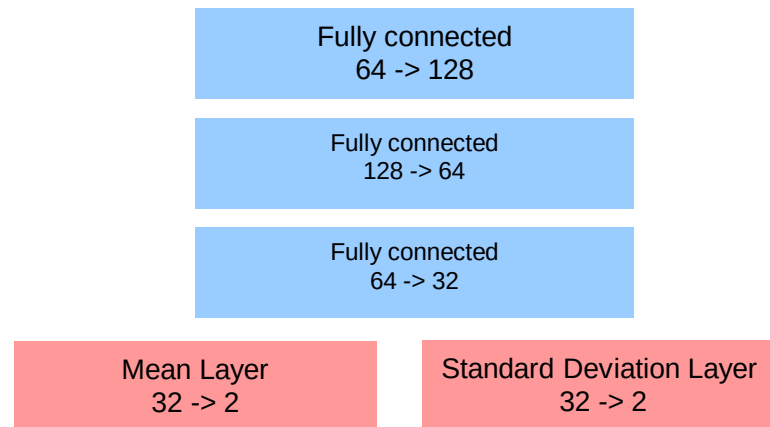
$$R_t = \lambda * \left(1 - \lambda(|z_t| - r_{circlce})^2\right) * \|z_t - z_{t-k}\|_2^2 + \lambda^2 * R_{t,bump}$$

Distillation

Each Distillation Dataset contains 250.000 samples, consist in:

- Observation state;
- Action distribution's mean;
- Action distribution's standard deviation.

Student Model



Metrics and Normalization

Loss Functions:

Mean Squared Error

$$L_{MSE}(x, y) = E[||x - y||_2^2]$$

Kullback-Leibler Divergence

$$L_{KL,\tau}(p|q) = E[\text{softmax}\left(\frac{p}{\tau}\right) \ln \left(\frac{\text{softmax}\left(\frac{p}{\tau}\right)}{\text{softmax}(q)}\right)]$$

Normalization to evaluate the students' performance:

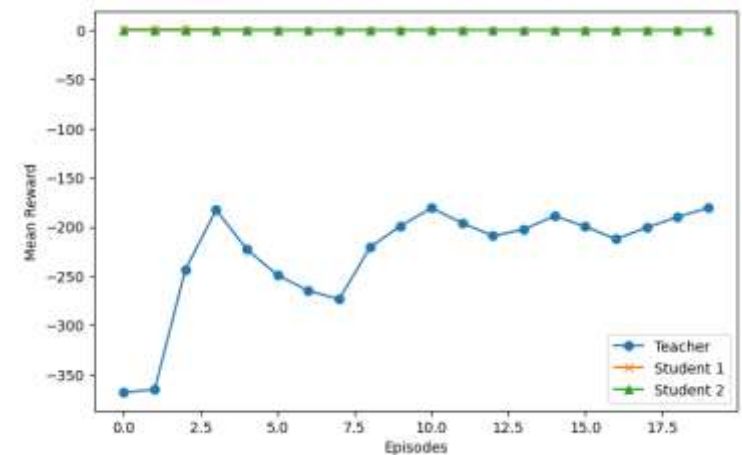
$$\text{Normalized Performance} = \frac{R_{student} - R_{teacher}}{|R_{teacher}|}$$

Results

Mean Squared Error Loss

Loss on test set	1° task	2° task
	0.3205	0.00004

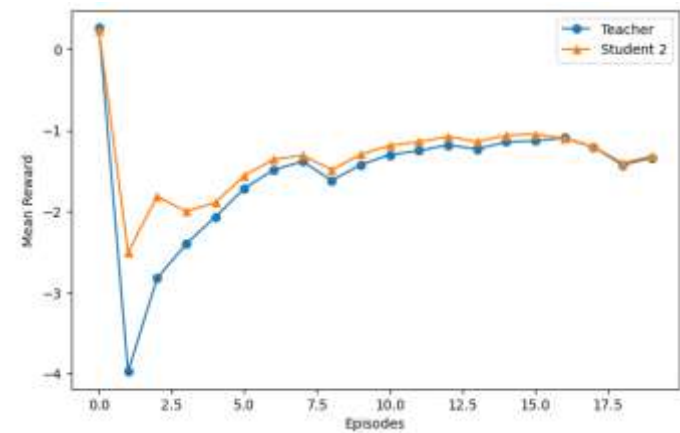
	Task 1- Reward	Task 1- Normalized Performance	Task 2 - Reward	Task 2 - Normalized Performance
Teacher Task 1	-180.56	0		
Teacher Task 2			-1.31	0
Student after Task 1	-0.099	0.9995		
Student after Task 2	0.008	1.0001	-1.33	-0.0153



Results

Mean Squared Error Loss

	Task 1 - Reward	Task 1 - Normalized Performance	Task 2 - Reward	Task 2 - Normalized Performance
Teacher Task 1	-180.56	0		
Teacher Task 2			-1.31	0
Student after Task 1	-0.099	0.9995		
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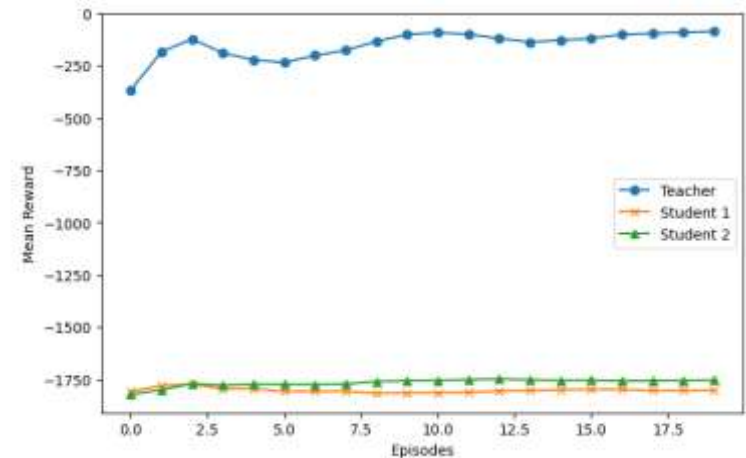


Results

Kullback-Leibler Divergence, $\tau = 0.1$

Loss on test set	1° task	2° task
	0.0009	0.0166

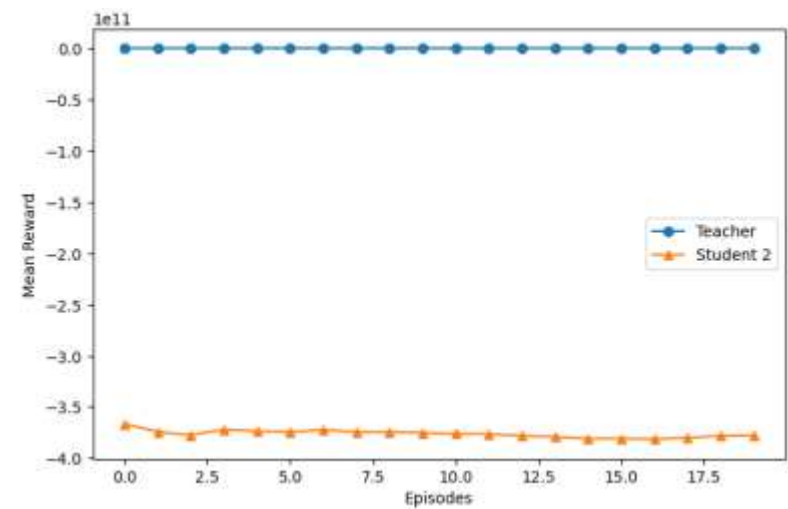
	Task 1- Reward	Task 1- Normalized Performance	Task 2 - Reward	Task 2 - Normalized Performance
Teacher Task 1	-86,12	0		
Teacher Task 2			-1,53	0
Student after Task 1	-1801.98	-19.924		
Student after Task 2	-1753,81	-19.3647	$-3,7 \cdot 10^{10}$	$-2,4 \cdot 10^{10}$



Results

Kullback-Leibler Divergence, $\tau = 0.1$

	Task 1- Reward	Task 1- Normalized Performance	Task 2 - Reward	Task 2 - Normalized Performance
Teacher Task 1	-86,12	0		
Teacher Task 2			-1,53	0
Student after Task 1	-1801.98	-19.924		
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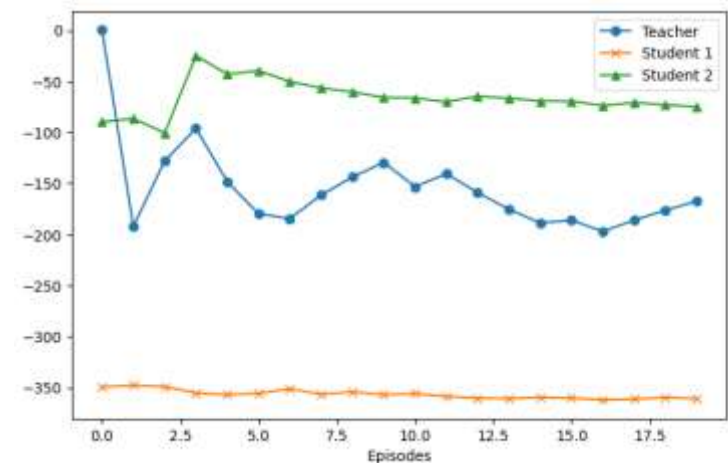


Results

Kullback-Leibler Divergence, $\tau = 1$

Loss on test set	1° task	2° task
	1.1354	0.0002

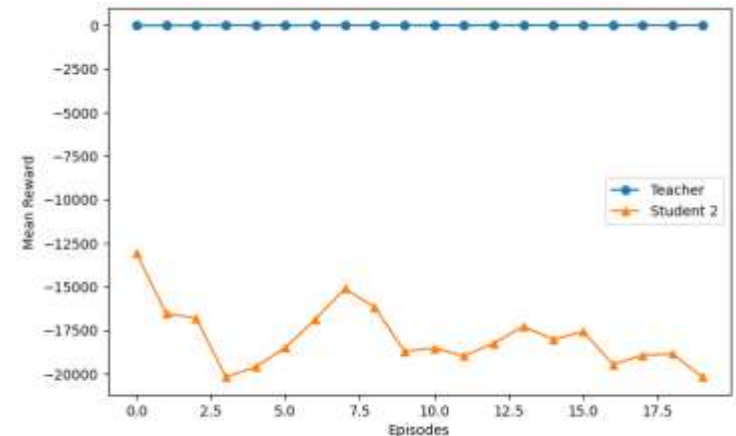
	Task 1- Reward	Task 1- Normalized Performance	Task 2 - Reward	Task 2 - Normalized Performance
Teacher Task 1	-167.46	0		
Teacher Task 2			-3.38	0
Student after Task 1	-360.85	-1.15		
Student after Task 2	-74.96	0.55	-20186.78	-5971.42



Results

Kullback-Leibler Divergence, $\tau = 1$

	Task 1 - Reward	Task 1 - Normalized Performance	Task 2 - Reward	Task 2 - Normalized Performance
Teacher Task 1	-167.46	0		
Teacher Task 2			-3.38	0
Student after Task 1	-360.85	-1.15		
Student after Task 2	-74.96	0.55	-20186.78	-5971.42



Results

Comparison

	Normalized Performance – 1° task	Normalized Performance – 1° task (after Task 2)	Normalized Performance – 2° task
MSE Loss	0.9995	1.0001	-0.0153
KL Divergance – $\tau = 0.1$	-19.924	-19.3674	$-2.4 \cdot 10^{10}$
KL Divergance – $\tau = 1$	-1.15	0.55	-5971.42

Thank you for the attention!